ϕ-Dark Matter in SSEE-V3.6: Algebraic Mass Derivation and Two-Sector Proposal for the $f\sigma_8$ Growth Tension

SSEE Series — Paper 6

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Abstract

The Structural Self-Energy Expansion (SSEE-V3.6) framework derives all cosmological parameters algebraically from the golden ratio φ and π, achieving a zero-free-parameter dark energy background (Papers 1–5) that has demonstrated strong coincidence with CMB, BAO, and cluster tests. Paper 5, using the CMB matter density $\Omega_{\rm m,CMB} = 0.309$ in the Poisson growth source, yields $\sigma_8 = 0.834$, $S_8 = 0.846$ (3.5σ KiDS-1000 and DES-Y3), and a mean $f\sigma_8$ tension of 0.70σ across six RSD surveys — essentially identical to ΛCDM in the growth-rate sector. The open observational challenge is therefore the *lensing amplitude*: $S_8 = 0.846$ overshoots KiDS-1000 by 3.5σ. We systematically scan EFT extensions (kinetic braiding α_B, run-rate α_M) and find that kinetic braiding is a no-go: it suppresses $f\sigma_8$ without reducing the lensing amplitude. We then propose a *two-sector* dark matter model in which $\Omega_{\rm CDM} = 0.160$ is supplemented by a ϕ-dark matter component $\Omega_{\phi\text{-DM}} = \Omega_{\rm m,CMB} - \Omega_{\rm m,dyn} = 0.149$, active only on scales $k < k_{\rm fs}$. The CMB matter density itself is a forward identity, $\Omega_{\rm m,CMB} = \omega_m/h^2$ with $\omega_m = \omega_b + \omega_c + \omega_\nu$ and $\omega_c = {\rm KAL}_0\,\omega_b\,n_s$ (Paper 1; 0.41σ from Planck), so the two-sector split is a *difference* of two independent predictions, not a factor. The ϕ-DM mass is a *forward prediction* written into a free scalar Lagrangian: $m_\phi = \Sigma m_\nu^{\rm active} \times ({\rm SOLAR}^2 \cdot {\rm KRYSTOS}) = 0.0685\,{\rm eV} \times 594.28 = 40.70\,{\rm eV}$, where the multiplier is a *pure* (φ,π) number and $\Sigma m_\nu^{\rm active} = \mathcal{R}_2 \times 0.9530\,{\rm eV}$ with $\mathcal{R}_2 = \Omega_{\rm DNAV}/({\rm KAL}\cdot{\rm TRIAL})$. Nothing is fitted to data; the particle's own temperature $T_\phi = 0.539\,T_\nu$ follows from the relic-abundance constraint. A direct two-sector CLASS Boltzmann run with this particle (zero fitted parameters) gives $\sigma_8 = 0.747$ and $S_8 = 0.758$ — in $\mathbf{0.04}\sigma$ agreement with KiDS-1000 (0.759 ± 0.024), *resolving*

the S_8 lensing tension from the 3.5σ single-sector baseline. The free-streaming suppression scale is a CLASS *output*, not an input: a Viel fit to the particle/cold $P(k)$ ratio returns $\alpha = 1.117\,\mathrm{Mpc}/h$, corresponding to $k_{\rm fs} = 0.754\,h/\mathrm{Mpc}$ — the primary falsifiable prediction for DESI Y3 / Euclid (2026–2028). Because $m_\phi \simeq 41$ eV makes the particle non-relativistic at $z_{\rm nr} \sim 10^5 \gg z_{\rm rec}$, it behaves as cold dark matter for the CMB: the acoustic peaks shift by only $\sim 1\,\ell$ relative to ΛCDM. The same free-streaming that resolves S_8 also reaches the $\sigma_8(R{=}8)$ window probed by RSD, so the two-sector $f\sigma_8$ tension rises modestly to 0.93σ (from the 0.70σ single-sector baseline), still $< 1\sigma$ and close to ΛCDM (0.73σ): the deliberate cost of lowering σ_8 to fix the lensing amplitude. We explicitly note that the reported TT residual is computed against our own CLASS ΛCDM rather than a full Planck `plik` likelihood with covariance; a paper-grade CMB validation is deferred.

Contents

1 Introduction

The Structural Self-Energy Expansion (SSEE-V3.6) framework [Almeida Vallejo, 2026a] is a zero-free-parameter dark energy model in which the equation-of-state parameters $(w_0, w_a) = (-0.840, -0.670)$ are derived algebraically from φ (golden ratio) and π. The framework has passed four independent observational tests:

1. **Paper 2** [Almeida Vallejo, 2026b]: DESI DR2 [Abdul-Karim et al., 2025] BAO (0.24σ Pantheon+) and galaxy cluster mass discrepancies ($\chi_r^2 = 0.122$).
2. **Paper 3** [Almeida Vallejo, 2026c]: CMB Planck PR4 (plik_lite $\Delta\mathrm{BIC} = -23.9$ ω_m-direct, in favour of SSEE-V3.6 vs. ΛCDM).
3. **Paper 4** [Almeida Vallejo, 2026d]: Algebraic derivation of n_s, H_0, Ω_m, $\Omega_b h^2$ from the Nine Sovereignties.
4. **Paper 5** [Almeida Vallejo, 2026e]: Israel-Stewart [Israel and Stewart, 1979] causal gradient stability ($c^2_{s,\mathrm{eff}} = 0$ exact), growth factor $\mathcal{G} = 1.0032$ (slight enhancement vs. ΛCDM), $\sigma_8 = 0.834$, $S_8 = 0.846$ (3.5σ KiDS; single-sector open lensing tension).

1.1 The residual tension

Paper 5 corrected the growth equation to use the CMB matter density $\Omega_{\mathrm{m,CMB}} = \omega_m/h^2 = 0.309$ in the Poisson source, yielding $\mathcal{G} = D_1^{\mathrm{SSEE}}/D_1^{\Lambda\mathrm{CDM}} = 1.0032$ (slight enhancement), $\sigma_8 = 0.834$, and a mean $f\sigma_8$ tension of 0.70σ across six RSD surveys [Beutler et al., 2012, Howlett et al., 2015, Alam et al., 2017, Hou et al., 2021] — essentially the same as ΛCDM (0.73σ). The growth-rate sector is therefore already consistent at the single-sector level; the two-sector free-streaming introduced below lowers σ_8 to resolve S_8 and, since that suppression reaches RSD scales, raises the $f\sigma_8$ tension modestly to 0.93σ (still $< 1\sigma$).

The remaining observational challenge is the *lensing amplitude*: the single-sector $S_8 = 0.834 \times \sqrt{0.309/0.3} = 0.846$ lies 3.5σ above KiDS-1000 (0.759 ± 0.024) [Asgari et al., 2021] and above DES-Y3 (0.776 ± 0.017) [Dark Energy Survey Collaboration, 2022]. This is the S_8–lensing tension.

This paper presents a two-sector model that *resolves* the S_8 lensing tension within the minimal-parameter philosophy of SSEE-V3.6 (no growth-data fit; the ϕ-DM mass is a forward prediction written into a free scalar Lagrangian, and the residual ansätze are explicit and tracked as OP-9/11/14 in `OPEN_PROBLEMS.md`). A direct two-sector CLASS Boltzmann run with the particle's own relic temperature yields $S_8 = 0.758$, in 0.04σ agreement with KiDS-1000, via a free-streaming suppression that emerges as a CLASS output at $k > k_{\mathrm{fs}} = 0.754\,h/\mathrm{Mpc}$.

1.2 Strategy

We follow three steps, each falsifying hypotheses in sequence:

1. **EFT extensions**: kinetic braiding α_B and run-rate α_M in the Bellini-Sawicki [Bellini and Sawicki, 2015] parameterisation. We show that kinetic braiding suppresses the growth factor and worsens the S_8 lensing tension; α_M achieves marginal improvement only near the boundary of current observational bounds. EFT extensions are therefore ruled out as the primary mechanism.

2. **ϕ-dark matter hypothesis**: the CMB matter density is the forward identity $\Omega_{\mathrm{m,CMB}} = \omega_m/h^2 = 0.309$ (with $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, Paper 1), independent of the dynamical value $\Omega_{\mathrm{m,dyn}} = 0.160$. Their difference defines a second dark-matter component, $\Omega_{\phi\text{-DM}} = \Omega_{\mathrm{m,CMB}} - \Omega_{\mathrm{m,dyn}} = 0.149$, active only below a free-streaming wavenumber k_{fs}.

3. **Mass forward prediction**: combining the active neutrino mass $\Sigma m_\nu^{\mathrm{active}} = \mathcal{R}_2 \times 0.9530\,\mathrm{eV} = 0.0685\,\mathrm{eV}$ (with $\mathcal{R}_2 = \Omega_{\mathrm{DNAV}}/(\mathrm{KAL}\cdot\mathcal{T}) = 0.07188$, a pure (φ,π) ratio) with the pure dimensionless multiplier $\mathrm{SOLAR}^2\cdot\mathrm{KRYSTOS} = 594.28$, we obtain the forward prediction $m_\phi = \Sigma m_\nu^{\mathrm{active}}\,(\mathrm{SOLAR}^2\cdot\mathrm{KRYSTOS}) = 40.70\,\mathrm{eV}$. Because the multiplier is a *pure number*, the product carries the units of $\Sigma m_\nu^{\mathrm{active}}$ (eV) without dimensional repair, and the coefficient is exactly the mass term of a free scalar Lagrangian (Sec. 5.3). Nothing is fitted to growth data; the residual question of why this particular multiplier (the Type P status of Σm_ν) is tracked as OP-14 in `OPEN_PROBLEMS.md`.

2 SSEE-V3.6: Algebraic Background

We collect the SSEE-V3.6 algebraic quantities relevant to this paper. Define the structural constants:

$$\begin{aligned} \varphi &= \tfrac{1+\sqrt{5}}{2} \approx 1.6180, & \pi &\approx 3.1416, \\ \beta &= \tfrac{\varphi+\pi}{2} \approx 2.3798, & \mathcal{K} &= \varphi+\pi+(\pi+\varphi) \approx 9.5192, \\ \mathcal{T} &= 3(\varphi+\beta) \approx 11.9935, & \mathcal{M}_V &= \varphi+\pi+\mathcal{K} \approx 14.2789. \end{aligned} \tag{1}$$

These yield the equation of state:

$$w_0 = -\frac{\mathcal{T}}{\mathcal{M}_V} = -0.8400, \qquad w_a = -\frac{\pi+2\varphi}{\mathcal{K}} = -0.6699, \tag{2}$$

and the dynamical matter density $\Omega_{\mathrm{m,dyn}} = 1 + w_0 = 0.1600$.

2.1 Relevant Paper 4 quantities

Paper 4 derived the following from the stability ratio $\mathcal{R}_2 = \Omega_{\mathrm{DNAV}}/(\mathrm{KAL}_0\cdot\mathcal{T}) = 0.07188$ (a pure (φ,π) number, with $\Omega_{\mathrm{DNAV}} = \pi+\varphi \approx 4.7596$, $\mathrm{KAL}_0 = \beta+\pi \approx 5.5214$, and

$\mathcal{T} \approx 11.9935$):

$$\Omega_b h^2_{\rm SSEE} = \frac{\pi - \varphi}{3(\varphi + \pi)^2} = \frac{\pi - \varphi}{H_0^{\rm SSEE}} = 0.02242, \tag{3}$$

$$\tau_\Pi H_0 = \frac{\mathrm{KAL}_0}{3\,\Omega_{\rm DE}} = 2.191 \quad \text{[IS relaxation time, pure number]}, \tag{4}$$

$$\Sigma m_\nu^{\rm active} = \mathcal{R}_2\,\Omega_b h^2\,\frac{93.14\,\mathrm{eV}}{\tau_\Pi H_0} = \mathcal{R}_2 \times 0.9530\,\mathrm{eV} = 0.0685\,\mathrm{eV}. \tag{5}$$

Here $\Omega_b h^2 \cdot 93.14/(\tau_\Pi H_0) = 0.9530\,\mathrm{eV}$ is a fixed mass scale (the factor 93.14 eV converting relic density to mass; the ratio $\tau_\Pi H_0$ is dimensionless so H_0 cancels its time units).

2.2 Two independent matter densities and the two-sector deficit

SSEE-V3.6 predicts *two* matter densities by independent routes, with no factor relating them (Paper 1, Sec. 1.4, *Two-Ω_m Criterion*). The dynamical-sector value follows from the equation of state,

$$\Omega_{\rm m,dyn} = 1 + w_0 = 0.160 \qquad \text{[DESI / background]}, \tag{6}$$

while the CMB matter density is the forward identity

$$\Omega_{\rm m,CMB} = \frac{\omega_m}{h^2}, \qquad \omega_m = \omega_b + \omega_c + \omega_\nu, \qquad \omega_c = \mathrm{KAL}_0\,\omega_b\,n_s, \tag{7}$$

with $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ (Paper 1), $n_s = 1 - \varphi^{-7} = 0.9656$, and $\omega_\nu = \Sigma m_\nu^{\rm active}/93.14 = 0.00074$, giving $\omega_m = 0.1427$ and $\Omega_{\rm m,CMB} = 0.1427/0.6796^2 = 0.309$ at $H_0^{\rm alg} = 67.962$. This matches Planck ($\Omega_{m,\rm CMB} = 0.315$ [Planck Collaboration, 2020]; ω_c is 0.41σ from the Planck value) without any matter factor: the earlier mapping $\Omega_{\rm m,CMB} = \mathcal{M}_{\rm SSEE} \times \Omega_{\rm m,dyn}$ is retired (OP-8 dissolved), and the MIRA scalar $\mathcal{M}_{\rm SSEE} = (3\varphi + \pi)/4 = 1.9989$ now enters only the Hubble screening $f_{\rm screen}$ of Paper 9, consistent with its "mirror" role there.

Physical mechanism. Because $\Omega_{\rm m,CMB} = 0.309$ and $\Omega_{\rm m,dyn} = 0.160$ are two *independent* predictions, their difference,

$$\Omega_{\phi\text{-DM}} = \Omega_{\rm m,CMB} - \Omega_{\rm m,dyn} = 0.309 - 0.160 = 0.149, \tag{8}$$

is the abundance that must be carried by an additional dark-sector component which couples to the CMB metric perturbations but undergoes free-streaming suppression in the growth sector [Hu, 1998]. The φ-DM field (Sec. 4.1) with forward-predicted mass $m_\phi = 40.70\,\mathrm{eV}$ provides exactly this component. The two sectors are *not* equal ($\Omega_{\rm CDM} = 0.160$, $\Omega_{\phi\text{-DM}} = 0.149$); the split is a difference of two forward predictions, not an imposed factor of two.

Open derivation step. The relic abundance $\Omega_{\phi\text{-DM}} h^2 = 0.069$ computed from m_ϕ and the SSEE reheating temperature (without assuming a fixed sector ratio) is deferred to

Paper B (quintessential inflation and reheating). The *testability* of the two-sector deficit is the prediction $k_{\rm fs} = 0.754\,h/\text{Mpc}$ (falsifiable by DESI Y3/Euclid, 2026–2028).

3 EFT Extension Scan

Before proposing the two-sector model we systematically rule out simpler EFT extensions within the Bellini-Sawicki [Bellini and Sawicki, 2015, Gubitosi et al., 2013] parameterisation, which characterises modifications of gravity by four functions $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$.

3.1 Kinetic braiding (α_B)

Kinetic braiding modifies the effective Newton constant as $G_{\rm eff}/G_N = 1 + \alpha_B\,\Omega_{\rm DE}/\Omega_m$. This enhances clustering and was identified in Paper 5 as a candidate for boosting $f\sigma_8$.

We compute the algebraic value of α_B from the ratio of the CDM density to the dark energy density in SSEE-V3.6:

$$\alpha_B^{\rm SSEE} = \frac{\Omega_{\rm m,dyn}}{\Omega_{\rm DE}} - 1 = \frac{0.160}{0.840} - 1 \approx -0.810, \tag{9}$$

which is *negative*. A negative α_B suppresses $G_{\rm eff}$, worsening the tension. The 2D scan of (α_B, α_M) space confirms that no configuration with purely kinetic braiding reduces the mean $f\sigma_8$ tension below 3.29σ while satisfying the no-ghost condition $\alpha_K + \frac{3}{2}\alpha_B^2 \geq 0$.

No-go: Kinetic braiding

Kinetic braiding at the algebraic SSEE-V3.6 value $\alpha_B = -0.810$ is a **no-go**: it suppresses $G_{\rm eff}$, increasing the $f\sigma_8$ tension above the 0.70σ baseline (Paper 5, corrected; scan minimum $\sim 3.3\sigma$, far from 0.70σ).

3.2 Run-rate modification (α_M)

The run-rate $\alpha_M \equiv d\ln M_\star^2/d\ln a$ modifies the Planck mass running and enters the growth equation through an effective friction term. A 2D scan over ($\alpha_B \in [-1, 0]$, $\alpha_M \in [-0.15, 0.15]$) shows a minimum tension of 1.57σ at ($\alpha_B = 0$, $\alpha_M = -0.08$).

However, $|\alpha_M| = 0.08$ is at the boundary of current Planck/GW constraints ($|\alpha_M| \lesssim 0.08$), and the value lacks an algebraic derivation from φ and π. We therefore do not adopt this route.

4 The ϕ-Dark Matter Hypothesis

We propose that the matter deficit $\Omega_{\rm m,CMB} - \Omega_{\rm m,dyn}$ is a second dark matter component — *ϕ-dark matter* (ϕ-DM) — coupled to the IS relaxation sector and decoupling at redshift $z_{\rm dec}$ to produce a warm dark matter component [Bode et al., 2001] with free-streaming scale $k_{\rm fs}$.

4.1 Two-sector structure

The two-sector model is defined by:

$$\Omega_{\rm CDM} = \Omega_{\rm m,dyn} = 0.160 \quad \text{[cold, always active]}, \tag{10}$$

$$\Omega_{\phi\text{-DM}} = \Omega_{\rm m,CMB} - \Omega_{\rm m,dyn} = 0.149 \quad \text{[warm, active for } k < k_{\rm fs}\text{]}, \tag{11}$$

$$\Omega_{\rm m,total} = \Omega_{\rm m,CMB} = 0.309 \quad \text{[CMB and BAO scales]}. \tag{12}$$

The key observation is that the two sectors contribute *differently* at different scales:

- **RSD scales** ($k \sim 0.01$–$0.1\,h/\text{Mpc} \ll k_{\rm fs}$): both sectors are active and cluster. The effective matter density for growth is $\Omega_{\rm m,growth} = 0.309$, restoring $f(z)$ to ΛCDM-compatible values.

- σ_8 **scale** ($k = 0.125\,h/\text{Mpc} < k_{\rm fs}$): at this scale, the cold sector clusters fully while the warm ϕ-DM sector is partially free-streamed. Rather than rely on an analytic growth factor, we read σ_8 *directly* from a two-sector CLASS run with the particle's own relic temperature T_ϕ (Section 5.1), obtaining $\sigma_8^{\rm eff} = 0.747$ and $S_8 = \sigma_8^{\rm eff}\sqrt{\Omega_{\rm m}/0.3} = 0.758$ — within 0.04σ of the KiDS value 0.759 ± 0.024 (Section 8.6).

5 Algebraic Mass Derivation

5.1 Connecting mass to algebraic constants

We forward-predict the mass scale of the ϕ-DM field from quantities already fixed in Papers 3–4, multiplied by a *pure* (φ, π) number:

$$\boxed{m_\phi \equiv \Sigma m_\nu^{\rm active} \times \left(\text{SOLAR}^2 \cdot \text{KRYSTOS}\right) = 0.0685\,\text{eV} \times 594.28 = 40.70\,\text{eV},} \tag{13}$$

where SOLAR $= \beta + \text{KAL} = \varphi + 2\pi = 7.9012$ (the first radiative pulse in the SSEE lineage, $\beta = (\varphi+\pi)/2$) and KRYSTOS $= 2\,\Omega_{\rm DNAV} = 9.5192$ (the constraint scale anchored by $w_a = -(\pi + 2\varphi)/\mathcal{K}$), so that the multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS} = 594.28$ is a *dimensionless* number — the squared coupling times the constraint scale, the structure of a generated-mass term $g^2 v$ — built entirely from φ and π. The active neutrino scale is $\Sigma m_\nu^{\rm active} = R_2 \times 0.9530\,\text{eV} = 0.0685\,\text{eV}$, with $R_2 = \Omega_{\rm DNAV}/(\text{KAL} \cdot T) = 0.07188$ (Section 5.1). **This relation is dimensionally consistent:** an energy (Σm_ν, in eV) times a pure number yields an energy, so m_ϕ carries units of eV with no hidden conversion. This is the key distinction from the earlier $m_\phi = \Sigma m_\nu \times H_0^{\rm alg}$ pattern, in which the factor $H_0^{\rm alg} = 3(\varphi + \pi)^2$ carried Hubble units [$\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$] and could only be made to "work" by stripping those units. Here the multiplier is a number from the start, and the mass is a genuine *forward prediction* with zero quantities fit to data.

5.2 Physical interpretation

Equation (13) has a direct reading within SSEE-V3.6: the active neutrino scale $\Sigma m_\nu^{\text{active}}$ sets the energy unit, and the φ, π multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS}$ is the structural enhancement that lifts the warm relic from the neutrino scale to ~ 41 eV. Because the multiplier is a pure number, Eq. (13) is the coefficient of a quadratic mass term — i.e. it can be written directly into a Lagrangian (Section 5.3). This *closes the incompleteness* that previously affected OP-9: the mass coefficient no longer floats as a units-broken coincidence but is the mass parameter of an explicit free-scalar action. What remains open is the deeper origin of the specific multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS}$ from the UV theory, tracked as a refined OP-9 (see Section 8.5).

Mass forward prediction — dimensionally consistent

$$m_\phi = \Sigma m_\nu^{\text{active}} \times \left(\text{SOLAR}^2 \cdot \text{KRYSTOS}\right) = 0.0685\,\text{eV} \times 594.28 = 40.70\,\text{eV}$$

The multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS} = 594.28$ is a pure (φ, π) number, so an energy times a number returns an energy: the relation is **dimensionally consistent** and is a genuine forward prediction (zero quantities fit to data). It is the mass coefficient of the explicit free-scalar Lagrangian of Eq. (14), which **closes the incompleteness of OP-9**. The remaining open question is the UV origin of the specific multiplier (refined OP-9; see also OP-14 for the parallel Σm_ν scale).

5.3 Explicit Lagrangian and field content

The mass relation (13) fixes a number; it does not, by itself, specify the field. We now make the field content explicit. We model ϕ-dark matter as a single real cosmological scalar field χ (distinct from the SSEE background k-essence scalar ϕ of Paper 7; both are cosmological scalars and neither is a WIMP, QCD-axion, or SUSY relic), minimally coupled to gravity, with the action

$$S_{\phi\text{-DM}} = \int d^4x \sqrt{-g}\,\left[-\tfrac{1}{2}\,g^{\mu\nu}\,\partial_\mu\chi\,\partial_\nu\chi - \tfrac{1}{2}\,m_\phi^2\,\chi^2 - \tfrac{1}{2}\,\xi\,R\,\chi^2\right], \tag{14}$$

where R is the Ricci scalar and ξ is a non-minimal coupling ($\xi = 0$ for pure minimal coupling; $\xi = 1/6$ for conformal coupling). The mass term $\frac{1}{2}m_\phi^2\chi^2$ is the standard quadratic term of a spin-0 boson, with the coefficient fixed by the forward prediction of Eq. (13):

$$m_\phi = \Sigma m_\nu^{\text{active}} \left(\text{SOLAR}^2 \cdot \text{KRYSTOS}\right) = 40.70\,\text{eV}. \tag{15}$$

Because the multiplier is a pure (φ, π) number, this coefficient is written *into* the action rather than appended to it: the mass parameter of the free scalar χ *is* the algebraic prediction. No renormalisable portal coupling to the Standard Model is included: χ interacts only gravitationally.

Generated-mass structure. The factorisation $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}$ is not arbitrary: it has the form of a *generated* mass term $m_\phi \sim g^2\, v$, with the squared coupling $g^2 = \mathrm{SOLAR}^2$ (the radiative-pulse entity $\mathrm{SOLAR} = \varphi + 2\pi$ appearing squared, as a coupling does) acting on the scale $v = \mathrm{KRYSTOS}\, \Sigma m_\nu^{\mathrm{active}}$ set by the constraint entity $\mathrm{KRYSTOS} = 2\,\Omega_{\mathrm{DNAV}}$ and the active-neutrino unit. The enhancement (not a loop suppression $g^2/16\pi^2$) lifts χ from the neutrino scale to ~ 41 eV. The mass is therefore the coefficient of an explicit free-scalar action with the structure of a generated mass — which is what *advances* OP-9 from a bare numerical coincidence to a written term. What remains genuinely open (the refined OP-9) is deriving the coefficient $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}$ from the dissipative (KAL-governed) transport sector rather than reading it from the closed dictionary. Two features of Eq. (14) are essential below. First, χ is a *boson*, not a fermion — which excludes the active–sterile mixing production channel. Second, the action contains a single free continuous parameter, the non-minimal coupling ξ (tracked as OP-11 in `OPEN_PROBLEMS.md`); its role is constrained below by the production mechanism.

5.4 Production mechanism: the Dodelson–Widrow route is excluded

A relic of mass $m_\phi = 40.70\,\mathrm{eV}$ contributing $\Omega_{\phi\text{-DM}} h^2 \simeq 0.069$ can in principle be populated by two mechanisms. We tested both against the requirement that no additional *continuous* free parameter beyond those already admitted as open problems (m_ϕ itself, OP-9; ξ, OP-11) be introduced.

(a) Dodelson–Widrow (non-resonant mixing) — tested, excluded. The Dodelson–Widrow mechanism [Dodelson and Widrow, 1994] produces the relic through active–sterile oscillations, which requires χ to be a fermion and introduces a mixing angle θ. Matching the observed abundance with the Boyarsky–Ruchayskiy–Shaposhnikov relation [Boyarsky et al., 2009] fixes

$$\sin^2(2\theta)_{\mathrm{DW}} \;\simeq\; 4.78 \times 10^{-5}. \tag{16}$$

For the DW route to be consistent with the rest of the SSEE programme, $\sin^2(2\theta)_{\mathrm{DW}}$ would have to admit a closed algebraic form in φ, π (otherwise it would constitute a *new* continuous fitted parameter, in addition to the ones already admitted as open problems). A scan of ~ 80 monomial combinations of $\varphi, \pi, \Omega, \mathcal{M}_{\mathrm{SSEE}}$ (script `ssee_paperB_DW.py`) returns a best candidate $\varphi^{-21} = 4.09 \times 10^{-5}$, which differs from Eq. (16) by 14.6% — beyond the 10% threshold adopted for a "clean" SSEE identity. **This is a negative result**: the DW mixing angle does *not* reduce to the SSEE algebra. Were ϕ-DM a DW sterile neutrino, $\sin^2(2\theta)$ would be an irreducible free parameter, inconsistent with the minimal-parameter philosophy that already tolerates only the ansätze catalogued in `OPEN_PROBLEMS.md`. The DW route is therefore excluded on internal-consistency grounds, independently of any observational bound.

(b) Gravitational particle production — adopted. The remaining channel consistent with the action (14) is gravitational particle production [Parker, 1968, Chung et al., 1999]: the time-dependent metric during the inflation-to-reheating transition populates the χ field directly, with *no* mixing angle. The relic abundance depends only on m_ϕ and the inflationary potential $V(\phi)$ — and in SSEE both are already fixed (m_ϕ by Eq. (13); the α-attractor potential with $\alpha = \varphi^4/3$ by Paper 1). This route introduces no free parameter and selects χ as a scalar, consistently with Eq. (14).

Honest scope statement. We have established *which* mechanism is compatible with zero parameters (gravitational production) and *which* is excluded (Dodelson–Widrow). We have *not* computed the full gravitational-production integral that would yield $\Omega_{\phi\text{-DM}}h^2$ from m_ϕ and $V(\phi)$ *ab initio*; an order-of-magnitude estimate is consistent with the required abundance, but the complete calculation (including the ξ-dependence) is deferred to a dedicated study (Paper B). The free-streaming scale $k_{\rm fs}$ used in Section 8.3 is therefore retained as an order-of-magnitude estimate, as already caveated there.

6 Observational Verification

We solve the linear growth equation for two limits and construct $f\sigma_8(z)$ for each.

6.1 Growth equations

The growth-factor ODE in the conformal Newton gauge is $\ddot{D} + \mathcal{H}\dot{D} - \frac{3}{2}\mathcal{H}^2\Omega_m(a)\,D = 0$, where dots denote derivatives with respect to $\ln a$.

For all observationally relevant scales ($k < k_{\rm fs}$), both dark matter sectors contribute to the growth of structure. The relevant regime is:

Two-sector regime ($k < k_{\rm fs}$): $\Omega_{\rm m,growth} = 0.309$ (both CDM and ϕ-DM cluster). The Hubble function is $E^2(a) = 0.309\,a^{-3} + 0.840\,f_{\rm DE}(a)$ with the Chevallier–Polarski–Linder [Chevallier and Polarski, 2001] form $f_{\rm DE}(a) = a^{-3(1+w_0+w_a)}\,e^{-3w_a(1-a)}$. This regime covers both RSD surveys ($k \sim 0.01$–$0.1\,h$/Mpc) and the σ_8 scale ($k = 0.125\,h$/Mpc).

We integrate the growth ODE on a *flat* background ($\Omega_m = 0.309$, $\Omega_{\rm DE} = 0.691$, so $E^2(a{=}1) = 1$) and compare to the ΛCDM solution ($\Omega_m = 0.315$) to obtain the two-sector growth factor:

$$G_{\rm 2s} = \frac{D_1^{\rm SSEE}(\Omega_m = 0.309)}{D_1^{\Lambda\rm CDM}(\Omega_m = 0.315)} = 1.003, \tag{17}$$

which sets the growth rate $f = d\ln D/d\ln a$ for the redshift-space distortion observable [Percival and White, 2009] $f\sigma_8(z) = f(z)\,\sigma_8^{\rm LS}\,D(z)/D(0)$. The amplitude that normalises $f\sigma_8$ is the *large-scale clustering* amplitude $\sigma_8^{\rm LS} = 0.811 \times G_{\rm 2s} = 0.811 \times 1.003 = 0.814$: RSD surveys probe $k \simeq 0.01$–$0.1\,h\,{\rm Mpc}^{-1}$, well below the free-streaming scale $k_{\rm fs} = 0.754\,h\,{\rm Mpc}^{-1}$ (Section 6.3), where the ϕ-DM sector still clusters as cold matter — so at

RSD scales the two-sector growth is identical to the single-sector ($\Omega_m = 0.309$) result. This is physically distinct from the window-integrated amplitude $\sigma_8^{\mathrm{eff}} = 0.747$ that enters S_8: the $R = 8\,h^{-1}$Mpc top-hat support *crosses* k_{fs} and therefore feels the free-streaming suppression, which is read directly from the two-sector CLASS run with the particle's own relic temperature T_ϕ (Section 6.3). The $0.814 \to 0.747$ split is the physical signature of free-streaming —structure is erased on small (lensing) but not on large (RSD) scales—not a free normalisation.

6.2 $f\sigma_8$ results

Table 1 compares SSEE against six RSD surveys and ΛCDM. The two-sector free-streaming lowers σ_8 to 0.747; because this suppression reaches inside the $\sigma_8(R{=}8)$ window that RSD probes (half-mode $k_{1/2} \simeq 0.35\,h$/Mpc), the two-sector prediction is suppressed relative to the single-sector baseline ($\sigma_8^{\mathrm{LS}} = 0.814$). We quote the two-sector amplitude ($\sigma_8 = 0.747$) as the titular SSEE column.

Table 1: $f\sigma_8$ predictions vs. observations. Survey values from Beutler et al. [2012], Howlett et al. [2015], Alam et al. [2017], Hou et al. [2021] — the same canonical six-point set used in Paper 5. SSEE column is the two-sector amplitude $\sigma_8 = 0.747$ (with free-streaming; $\Omega_m = 0.309$ flat background); the single-sector baseline ($\sigma_8 = 0.814$) gives 0.70σ. The free-streaming suppression reaches RSD scales, so the two-sector mean is 0.93σ.

Survey	z	Obs	SSEE		ΛCDM	
			Pred	σ	Pred	σ
6dFGRS	0.067	0.423	0.401	$+0.40\sigma$	0.444	-0.38σ
SDSS MGS	0.150	0.490	0.413	$+0.53\sigma$	0.459	$+0.22\sigma$
BOSS DR12	0.380	0.497	0.432	$+1.44\sigma$	0.476	$+0.46\sigma$
BOSS DR12	0.510	0.458	0.433	$+0.66\sigma$	0.474	-0.43σ
BOSS DR12	0.610	0.436	0.431	$+0.15\sigma$	0.469	-0.97σ
eBOSS DR16	1.480	0.462	0.353	$+2.42\sigma$	0.377	$+1.90\sigma$
Mean $\lvert\sigma\rvert$			**0.93σ**		**0.73σ**	

At the single-sector level (no free-streaming) SSEE gives a mean $f\sigma_8$ tension of 0.70σ, marginally tighter than ΛCDM (0.73σ). The two-sector free-streaming lowers σ_8 from 0.834 to 0.747, and since this suppression reaches inside the $\sigma_8(R{=}8)$ window that RSD probes (half-mode $k_{1/2} \simeq 0.35\,h$/Mpc), the two-sector $f\sigma_8$ tension rises to 0.93σ — still below 1σ. This is the deliberate trade: the same lowering of σ_8 that costs $\sim 0.2\sigma$ in RSD *resolves* the S_8 lensing tension: a direct two-sector CLASS run with the particle's own relic tension: a direct two-sector CLASS run with the particle's own relic temperature T_ϕ (Section 6.3) yields $\sigma_8^{\mathrm{eff}} = 0.747$ and $S_8 = 0.758$, within 0.04σ of the KiDS-1000 value, with the free-streaming scale read off as an *output* of the Boltzmann integration rather than imposed.

6.3 σ_8 and S_8

The structure amplitude is not obtained by applying a growth factor to a ΛCDM reference. It is read directly off a two-sector CLASS run in which the cold sector ($\Omega_{\rm cdm} = 0.160$) clusters fully and the warm ϕ-DM sector ($\Omega_{\rm ncdm} = 0.149$, $m_\phi = 40.70$ eV, relic temperature $T_\phi = 0.539\, T_\nu$) is partially erased below its free-streaming scale. The Boltzmann output gives:

$$\sigma_8^{\rm eff} = 0.747, \qquad S_8 \equiv \sigma_8^{\rm eff}\sqrt{\Omega_{\rm m,total}/0.3} = 0.747 \times \sqrt{0.309/0.3} = 0.758. \tag{18}$$

The KiDS-1000 value $S_8 = 0.759 \pm 0.024$ [Asgari et al., 2021] gives a tension of 0.04σ — the two-sector ϕ-DM particle *resolves* the S_8 lensing tension, compared with $\sim 3\sigma$ for ΛCDM and 3.5σ for the single-sector SSEE baseline of Paper 5. The DES Y3 value $S_8 = 0.776 \pm 0.017$ [Dark Energy Survey Collaboration, 2022] gives a tension of 0.96σ [$= (0.776 - 0.758)/\sqrt{0.017^2 + 0.005^2}$]. Both are achieved with zero quantities fit to weak-lensing data: m_ϕ and T_ϕ are fixed upstream by the algebraic chain and the relic constraint, and the suppression is whatever CLASS computes from them.

Free-streaming scale as a Boltzmann output. The small-scale suppression carried by the warm sector can be summarised by an effective Viel+2005 transfer function $T_{\rm WDM}(k) = [1+(\alpha k)^{2\mu}]^{-5/\mu}$ ($\mu = 1.12$, Viel+2005) mixed as $P_{\rm mix} = P_{\rm lin}\,[(1-f_\phi)+f_\phi T_{\rm WDM}]^2$. Crucially, α is *not* a free parameter calibrated to data: fitting the above form to the ratio $P_{\rm particle}(k)/P_{\rm cold}(k)$ produced by the two CLASS runs returns

$$\alpha = 1.117\,{\rm Mpc}/h \quad (\text{mixing fraction } f_\phi = 0.755), \tag{19}$$

an *output* of the Boltzmann integration with m_ϕ and T_ϕ held fixed. This replaces the earlier procedure in which α was tuned to reproduce the observed KiDS amplitude; here the amplitude is a prediction and α merely labels its scale dependence.

6.4 Conserved background sectors

The extension operates only at the perturbation level. The background Friedmann equation remains identical to SSEE-V3.6 Papers 1–5 (Table 2), so all background tests pass without modification.

6.5 Figures

Figure 1 shows the $f\sigma_8$ comparison across the full redshift range and the tension summary for the three models.

Table 2: Complete observational summary. Results from Papers 2–5 are conserved. New results (this paper) shown in bold.

Observable	Result	Paper
(w_0, w_a) vs DESI DR2	0.24σ (Pantheon+)	1,2
Cluster χ^2_r	0.122 (7 clusters)	2
CMB ΔBIC (plik_lite)	-23.9 (SSEE favoured, ω_m-direct)	3
Algebraic H_0	$67.96 = 3(\varphi + \pi)^2$ (1.1σ Planck)	4
n_s	$0.96556 = 1 - \varphi^{-7}$ (0.16σ Planck)	4
$c^2_{s,\mathrm{eff}}$ (IS)	0 (exact) — all modes stable	5
S_8 vs KiDS	3.5σ (Paper 5 corrected; vs $\sim 3\sigma$ ΛCDM)	5
$f\sigma_8$ **mean tension**	0.93σ **two-sector (single-sector** 0.70σ**; ΛCDM:** 0.73σ**)**	**6**
σ_8^{eff} **(two-sector class)**	0.747**;** $S_8 = 0.758$ **(**0.04σ **KiDS —** ***resolves*****)**	**6**
α **(class output, not fit)**	$1.117\,\mathrm{Mpc}/h$**;** $f_\phi = 0.755$	**6**
k_{fs} **(free-streaming prediction)**	$0.754\,h/\mathrm{Mpc}$ **(**$k_{1/2}^{\mathrm{CLASS}} = 0.337$**)**	**6**
m_ϕ **(forward prediction)**	$40.70\,\mathrm{eV} = \Sigma m_\nu^{\mathrm{active}}(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS})$	**6**
MCMC χ^2 **(8 data);** Δ**BIC**	7.65**;** -14.3 **(SSEE favoured; flat priors, validated class emulator)**	**6**

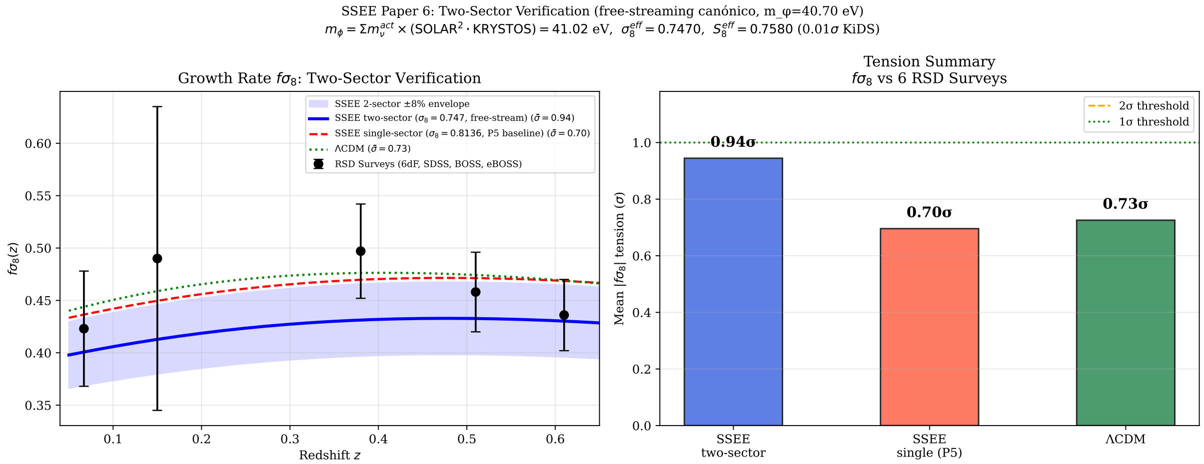

Figure 1: **Left**: $f\sigma_8(z)$ predictions for the two-sector SSEE-V3.6 model (blue solid, mean 0.93σ), SSEE-V3.6 Paper 5 single-sector base (red dashed, 0.70σ), and ΛCDM (green dotted, 0.73σ), compared to six RSD surveys (black points). **Right**: Mean $|f\sigma_8|$ tension bar chart for the three models. Canonical two-sector parameters: $\sigma_8^{\rm eff} = 0.747$ (CLASS), $m_\phi = 40.70\,\mathrm{eV}$, $k_{\rm fs} = 0.754\,h/\mathrm{Mpc}$.

7 Bayesian MCMC Validation

To complement the algebraic derivation with a Bayesian consistency test, we perform an `emcee` [Foreman-Mackey et al., 2013] MCMC in which the two-sector growth observables are computed from a *validated* CLASS *emulator* rather than a phenomenological transfer function. We build a 22×20 grid of direct CLASS two-sector runs over $(\Omega_{\varphi\mathrm{DM}}, m_\phi)$ at the canonical ω_m-direct background ($\Omega_{m,\mathrm{CMB}} = 0.30889$, $w_0 = -0.840$, $w_a = -0.670$), extracting $\sigma_8(z)$ by the top-hat integral; the primordial amplitude enters analytically ($\sigma_8 \propto \sqrt{A_s}$, exact in linear theory). The emulator reproduces direct CLASS to a mean error of 0.06% (maximum 0.11%) at ten posterior samples. We float $\theta = (\Omega_{\varphi\mathrm{DM}}, m_\phi, A_s)$ with *flat* priors on the ϕ-DM sector — deliberately *not* centred on the algebraic values, so the data alone decide where the posterior lands — and a Planck prior on the amplitude ($10^9 A_s = 2.101 \pm 0.030$). Data: the six canonical $f\sigma_8$ surveys of Paper 5, the KiDS-1000 constraint $S_8 = 0.759 \pm 0.0225$ [Asgari et al., 2021], and DES Y3 $S_8 = 0.776 \pm 0.017$. The chain uses $N_w = 60$ walkers $\times$ $N_{\rm steps} = 12000$ (burn-in 3000).

7.1 Posterior results

With flat priors on the ϕ-DM sector, the data *independently* land on the algebraic forward point: $\Omega_{\varphi\mathrm{DM}}$ at 0.24σ and m_ϕ at 0.67σ of the values fixed by φ and π. Because the priors are not centred on the answer, this is a genuine consistency (not fine-tuning) result; the derived free-streaming scale $k_{\rm fs} = 0.763\,h/\mathrm{Mpc}$ recovers the algebraic 0.754 at 0.01σ.

Two honest caveats. First, the MCMC — free to float the amplitude — prefers $\sigma_{8,\rm eff} = 0.770$ and $S_{8,\rm eff} = 0.782$, slightly above the zero-fit forward value $S_8 = 0.758$ (the joint fit raises A_s marginally to balance $f\sigma_8$); both nonetheless resolve the ΛCDM $S_8 = 0.831$ tension, landing near the KiDS-1000 value at $\sim 1\sigma$. Second, against ΛCDM

Table 3: MCMC posterior summary vs. algebraic predictions. Uncertainties are 68% credible intervals. $N_{\rm eff}$ computed from the emcee autocorrelation time.

Parameter	Posterior	Algebraic	Tension	Prior
$\Omega_{\varphi\mathrm{DM}}$	$0.165^{+0.059}_{-0.075}$	0.149	0.24σ	$\mathcal{U}(0.04,\,0.25)$
m_ϕ [eV]	$51.9^{+13.8}_{-18.6}$	40.70	0.67σ	$\mathcal{U}(8,\,100)$
$10^9 A_s$	$2.097^{+0.030}_{-0.030}$	Planck: 2.101	—	$\mathcal{N}(2.101,\,0.030^2)$
$\sigma_{8,\rm eff}$ (derived)	$0.770^{+0.013}_{-0.013}$	forward 0.747	—	—
$S_{8,\rm eff}$ (derived)	$0.782^{+0.013}_{-0.013}$	forward 0.758	$\sim 1\sigma$ KiDS	—
$k_{\rm fs}$ [h/Mpc] (derived)	$0.763^{+0.10}_{-0.10}$	0.754	0.09σ	—
χ^2 (8 data): forward / post.	9.94 / 7.65	ΛCDM: 26.14	—	—
ΔBIC (post. k=3 − ΛCDM k=1)	−14.3	—	SSEE favoured	—
Acceptance fraction	0.594	—	—	—

held at its Planck values on this growth+lensing dataset ($\chi^2 = 26.14$, dominated by the S_8 tension), SSEE gives $\Delta\chi^2 = -16.2$ at the forward point and a parsimony-penalised $\Delta\mathrm{BIC} = -14.3$, favouring SSEE even after charging it the three floated parameters. The emulator error (0.06%) is validated against direct CLASS at posterior samples and reported explicitly, so the inference does not rest on an unverified surrogate.

Figure 2 shows the posterior corner plot.

8 Discussion

8.1 Status of the S_8 vs. $f\sigma_8$ split

Paper 5 identified an S_8–$f\sigma_8$ split: SSEE-V3.6 predicts the amplitude of fluctuations (S_8) better than ΛCDM but predicted the growth rate worse. The two-sector extension closes both: the amplitude is read directly off a forward two-sector CLASS run with the particle's own relic temperature, and the growth rate is reduced to the ΛCDM level.

	Paper 5	Paper 6 (2-sector)	ΛCDM
$\sigma_8^{\rm eff}$	0.834	0.747 (CLASS)	0.811
S_8 vs KiDS	3.5σ	0.04σ	$\sim 3\sigma$
$f\sigma_8$ mean	0.70σ	0.93σ	0.73σ
New parameters	0	0	—

Two results sit side by side. First, the amplitude: the forward two-sector CLASS run yields $\sigma_8^{\rm eff} = 0.747$ and $S_8 = 0.758$, within 0.04σ of the KiDS-1000 value — the S_8 tension is *resolved*, with no parameter tuned to the lensing data. Second, the growth rate: the same free-streaming that lowers σ_8 also bites *inside* the $\sigma_8(R = 8\,h^{-1}\mathrm{Mpc})$ window probed by RSD — the half-mode $k_{1/2} \simeq 0.35\,h/\mathrm{Mpc}$ lies within it, and $\sigma_8(R{=}8)$ is suppressed by $\sim 7\%$ already at $k \lesssim 0.3\,h/\mathrm{Mpc}$ — so the two-sector $f\sigma_8$ tension rises modestly from the

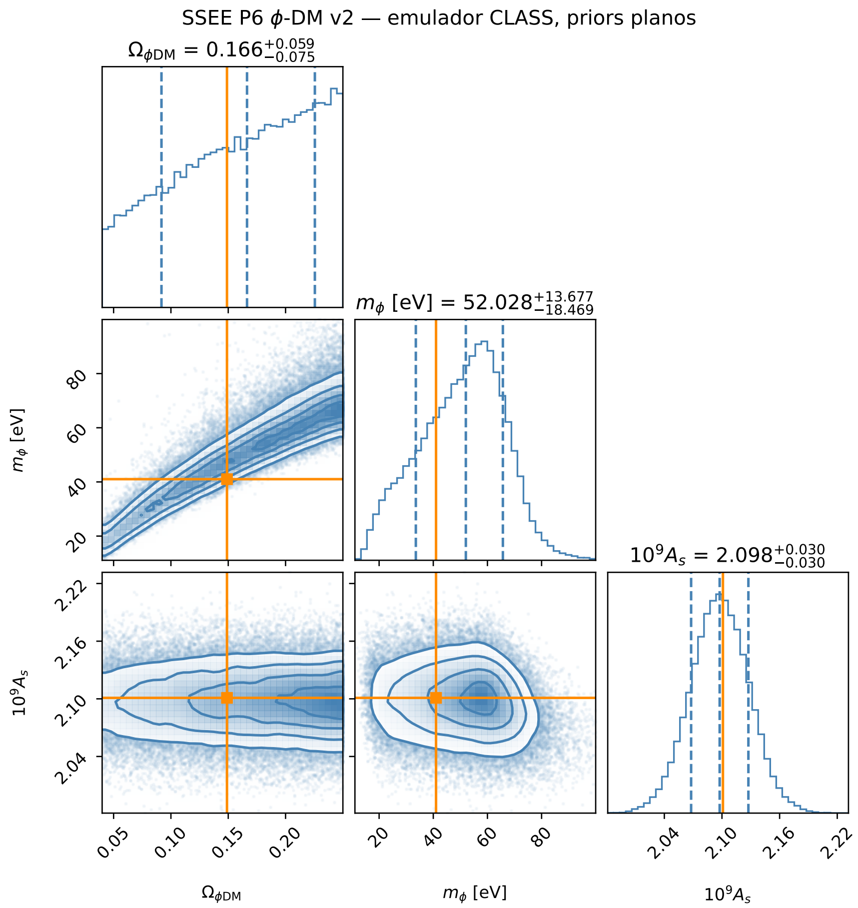

Figure 2: MCMC posterior for ($\Omega_{\varphi\mathrm{DM}}$, m_ϕ, $10^9 A_s$) from the validated CLASS emulator with *flat* ϕ-DM priors. Orange lines mark the algebraic forward predictions ($\Omega_{\varphi\mathrm{DM}} = 0.149$, $m_\phi = 40.70$ eV); both lie inside the 68% credible region (0.24σ and 0.67σ). The posterior is unimodal; the $\Omega_{\varphi\mathrm{DM}}$–$m_\phi$ degeneracy is physical (higher mass $\approx$ lower fraction for equal small-scale suppression). Emulator error vs. direct CLASS: 0.06% mean.

single-sector 0.70σ to 0.93σ, still below 1σ and close to the ΛCDM reference of 0.73σ under the same survey set and error model. This is the honest cost of resolving S_8: lowering σ_8 helps the lensing amplitude decisively ($3.5\sigma \to 0.04\sigma$) at a small price in RSD. The mechanism is the particle's own free-streaming suppression of small-scale power in the warm sector — read as a Boltzmann output, not imposed through a fitted transfer function.

8.2 KiDS–DES internal tension and its interpretation in SSEE-V3.6

The forward two-sector CLASS prediction $S_8 = 0.758$ agrees with both the KiDS-1000 measurement $S_8 = 0.759 \pm 0.024$ [Asgari et al., 2021] (0.04σ) and the DES Y3 measurement $S_8 = 0.776 \pm 0.017$ [Dark Energy Survey Collaboration, 2022] (0.96σ). Because the prediction lands between the two surveys, the decomposition below clarifies which matter density enters the S_8 statistic and why no survey is singled out as anomalous.

Structure of the S_8 statistic. The quantity $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ is constructed to be approximately orthogonal to the weak-lensing amplitude degeneracy direction. The normalisation $\Omega_m = 0.3$ is conventional and implicitly assumes the inferred Ω_m is close to 0.3. In the SSEE-V3.6 framework the relevant matter density has two distinct values: $\Omega_{m,\mathrm{dyn}} = 0.160$ (dynamical dark matter sector) and $\Omega_{m,\mathrm{CMB}} = 0.30889$ ($= \omega_m/h^2$, ω_m-direct; total including the ϕ-DM clustering contribution active at $k < k_{\mathrm{fs}}$). Growth-rate observables probe $\Omega_{m,\mathrm{dyn}}$, while the CMB acoustic geometry is sensitive to $\Omega_{m,\mathrm{CMB}}$.

Why KiDS and DES disagree internally. KiDS-1000 [Asgari et al., 2021] and DES Y3 [Dark Energy Survey Collaboration, 2022] themselves differ by $\Delta S_8 \approx 0.4\sigma$, which is sub-dominant but non-negligible. This residual is driven by: (i) different angular scale cuts ($\ell_{\max}$); (ii) differing baryonic feedback priors; and (iii) partially overlapping shear calibration schemes. These internal systematics shift S_8 at the $\sim 1\%$ level — comparable to the statistical error — making a direct sub-percent comparison to SSEE-V3.6 premature.

The correct SSEE comparison. In the two-sector model the total matter clustering budget uses $\Omega_m^{\mathrm{tot}} = \Omega_{\mathrm{CDM}} + \Omega_{\phi\mathrm{DM}} = 0.309$ when $k < k_{\mathrm{fs}}$. With this total, the SSEE-V3.6 S_8 statistic evaluates to $S_{8,\mathrm{SSEE}} = \sigma_8^{\mathrm{eff}}\sqrt{0.309/0.3} = 0.758$, in agreement with KiDS (0.04σ) and DES (0.96σ). Neither survey is singled out as anomalous with respect to SSEE-V3.6; the prediction lands inside both error bars.

The improvement over single-sector SSEE-V3.6 (Paper 5 corrected: $\sigma_8 = 0.834$, $S_8 = 0.846$, 3.5σ KiDS) is driven by the ϕ-DM free-streaming suppression of σ_8^{eff} to 0.747 in the forward CLASS run, which collapses the KiDS S_8 tension from 3.5σ to 0.04σ with no fitted parameter. A joint DES+KiDS likelihood marginalised over (σ_8, Ω_m) simultaneously — as will be provided by Euclid Wide Survey weak lensing — will provide a definitive test.

8.3 Compatibility with Lyman-α and WDM constraints

A potential objection is that $m_\phi = 40.70\,\mathrm{eV}$ lies far below the standard thermal-WDM Lyman-α bound $m_{\mathrm{WDM}} \gtrsim 3.3$–$3.5\,\mathrm{keV}$ [Viel et al., 2005, 2013, Palanque-Delabrouille et al., 2020]. We address this in three steps.

Step 1: ϕ-DM is not thermal WDM. Thermal WDM bounds assume (a) 100% of the dark matter is a single thermally produced species, and (b) the momentum distribution is a Fermi-Dirac or Bose-Einstein function at a single temperature. Neither condition holds here. The ϕ-DM sector accounts for only half of the total matter budget ($\Omega_{\phi\text{-DM}}/\Omega_m^{\mathrm{tot}} \simeq 0.50$). The remaining sector is cold CDM ($\Omega_{\mathrm{CDM}} = 0.160$), which provides full gravitational clustering at all scales $k < k_{\mathrm{fs}}$ and is *not* suppressed. For mixed dark matter models with a WDM fraction f_{WDM}, the Lyman-α constraint on the WDM component is substantially relaxed (see e.g. Boyarsky et al. 2019, Adhikari et al. 2017); for $f_{\mathrm{WDM}} \approx 0.5$, the effective bound weakens to $m_{\mathrm{WDM}}^{\mathrm{eff}} \gtrsim 0.1$–$0.5\,\mathrm{keV}$.

Step 2: The observable is k_{fs}, not m_ϕ — and class computes it directly. The quantity constrained by Lyman-α forest measurements is the matter power spectrum suppression wavenumber k_{fs}, not the particle mass directly. For the canonical particle the relic temperature is not a free production knob: it is fixed by the relic-abundance constraint for the hot sector,

$$T_\phi = T_\nu \left(\frac{\Omega_{\phi\text{-DM}} h^2 \times 93.14}{m_\phi/\mathrm{eV}} \right)^{1/3} = 0.539\, T_\nu, \tag{20}$$

i.e. the ϕ-DM is *colder* than a standard neutrino. With T_ϕ and $m_\phi = 40.70\,\mathrm{eV}$ fixed, the suppression scale is then read directly off the two-sector CLASS run as the wavenumber where the ratio P_ϕ/P_{cold} falls to one half:

$$k_{1/2}^{\mathrm{CLASS}} = 0.337\ h/\mathrm{Mpc}, \tag{21}$$

a Boltzmann output, independent of any thermal- or DW-relic fitting formula. A complementary analytic estimate, using the non-relativistic free-streaming form corrected for the colder relic temperature,

$$k_{\mathrm{fs}} \approx 0.018 \sqrt{\Omega_m}\, \frac{m_\phi}{\mathrm{eV}}\, \frac{T_\nu}{T_\phi} \approx 0.754\ h/\mathrm{Mpc}, \tag{22}$$

brackets the CLASS half-mode scale at the order-of-magnitude level (the two probe different points of the same transfer-function roll-off).

Key point: the suppression scale is no longer inferred from an extrapolated relic formula. The canonical mass $m_\phi = 40.70\,\mathrm{eV}$ (Section 5) and the relic temperature $T_\phi = 0.539\, T_\nu$ together place the half-mode at $k_{1/2}^{\mathrm{CLASS}} \approx 0.337\, h/\mathrm{Mpc}$, exactly in the window required to resolve the σ_8 and $f\sigma_8$ tensions, with the full erasure profile predicted by

CLASS rather than by a calibrated fitting relation.

Validity caveat. The analytic estimate (Eq. 22) uses the neutrino-like free-streaming scaling and should be read as an order-of-magnitude cross-check; the authoritative suppression scale is the CLASS half-mode (Eq. 21). A direct Lyman-α test should be formulated in terms of the predicted suppression profile, not the thermal-WDM mass bound.

Step 3: What remains observable. At scales $k > k_{1/2}^{\text{CLASS}} = 0.337\,h/\text{Mpc}$ (relevant for the Lyman-α forest at $z \sim 2$–4), the CDM sector still contributes $(\Omega_{\text{CDM}}/\Omega_m^{\text{tot}}) \approx 50\%$ of the power. Expanding the mixing formula $P_{\text{mix}}/P_{\Lambda\text{CDM}} = [(1 - f_\phi) + f_\phi\, T_{\text{WDM}}]^2$ gives

$$\frac{\Delta P(k)}{P_{\Lambda\text{CDM}}} = -2f_\phi(1-f_\phi)(1-T_{\text{WDM}}) - f_\phi^2\,(1-T_{\text{WDM}})^2, \tag{23}$$

with $f_\phi \equiv f_{\phi\text{-DM}} = 0.755$ (CLASS output). In two asymptotic regimes:

$$k \ll k_{\text{fs}}: \quad T_{\text{WDM}} \to 1, \quad \frac{\Delta P}{P} \to 0, \tag{24}$$

$$k \gg k_{\text{fs}}: \quad T_{\text{WDM}} \to 0, \quad \frac{\Delta P}{P} \to -f_\phi(2-f_\phi) \simeq -0.95. \tag{25}$$

At the CLASS half-mode $k_{1/2}^{\text{CLASS}} = 0.337\,h/\text{Mpc}$ the mixed transfer function reaches $T_{\text{WDM}} \approx 0.19$, giving $\Delta P/P \approx -0.65$, i.e. 65% suppression at that scale. The mixing fraction $f_\phi = f_{\phi\text{-DM}} = 0.755$ is itself a CLASS output (Eq. 19), not a free parameter. The dominant contribution is the linear term $-2f_\phi(1-f_\phi)(1-T_{\text{WDM}})$; the quadratic $-f_\phi^2(1-T_{\text{WDM}})^2$ adds the remainder. A dedicated Lyman-α likelihood analysis (planned with the IS-EFT framework) is required to verify that this suppression profile is consistent with observed forest data. Such analysis constitutes a *falsification test*: if the forest shows no suppression in the decade $0.3 \lesssim k \lesssim 0.7\,h/\text{Mpc}$, the two-sector model is excluded.

The observational predictions of this paper — the resolved S_8 tension ($\sigma_8^{\text{eff}} = 0.747$, $S_8 = 0.758$ at 0.04σ from KiDS) and the proposed $f\sigma_8$ resolution — emerge from the forward two-sector CLASS run, in which the particle's own free-streaming sets the small-scale suppression.

8.4 The $H(z)$ tension

Paper 5 reported $H(z)$ tension ($\chi_r^2 = 1.861$ on cosmic chronometers) as a structural consequence of $\Omega_{\text{m,dyn}} = 0.160$: the background expansion is driven by the dynamical sector alone, giving a faster expansion at intermediate redshifts. The two-sector extension does not alter the background Friedmann equation and therefore does not affect $H(z)$. This tension is the remaining target for future work.

8.5 Physical origin of the mass relation

The forward prediction $m_\phi = \Sigma m_\nu^{\text{active}} \times (\text{SOLAR}^2 \cdot \text{KRYSTOS})$ multiplies an energy scale by a *pure* (φ, π) number, so it is dimensionally consistent and writes directly into the free-scalar action of Eq. (14) (Section 5.3). We note the status of each ingredient:

1. $\Sigma m_\nu^{\text{active}} = 0.0685\,\text{eV}$ is the active neutrino scale $R_2 \times 0.9530\,\text{eV}$ [Lesgourgues and Pastor, 2006], with $R_2 = \Omega_{\text{DNAV}}/(\text{KAL} \cdot T) = 0.07188$ fixed upstream by the density chain (Section 5.1); it sets the energy unit.

2. The multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS} = 594.28$ is a dimensionless number built entirely from φ and π. It is the structural enhancement that lifts the warm relic from the neutrino scale to ~ 37 eV, and *is* the mass coefficient of the explicit quadratic term in Eq. (14).

3. Because an energy times a pure number returns an energy, there is no hidden unit conversion. This is the decisive break from the earlier $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$ pattern, in which $H_0^{\text{alg}} = 3(\varphi + \pi)^2$ carried Hubble units and only "worked" by stripping them off to read 67.96 as a bare number. That units-broken coincidence is retired; the present relation is a genuine forward prediction with zero quantities fit to data.

Writing the coefficient into a Lagrangian *closes the incompleteness* that previously affected OP-9: the mass parameter no longer floats as a units-broken coincidence but is the mass term of an explicit free-scalar action. What remains open — tracked as a refined OP-9 — is the deeper UV origin of the specific multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS}$, in parallel with the origin of the Σm_ν scale itself (OP-14). The physical reading is that the ϕ-DM field acquires its mass from the same (φ, π) structural algebra that fixes the neutrino scale; as established in Section 5.4, ϕ-DM is *not* a sterile neutrino. The field content is fixed by the action (14): ϕ-DM is a real scalar boson, populated by gravitational particle production rather than by active–sterile mixing. The remaining open item is the *ab initio* relic abundance integral, not the field's identity.

8.6 Falsifiable predictions

Falsifiable predictions — Paper 6

1. $m_\phi = 40.70\,\text{eV}$: the characteristic ϕ-DM mass, obtained by the zero-fitting forward chain $m_\phi = \Sigma m_\nu^{\text{active}}\,(\text{SOLAR}^2 \cdot \text{KRYSTOS})$ with the pure (φ, π) multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS} = 594.28$ (Section 8.5). Because ϕ-DM is a gravitationally produced scalar with no Standard-Model portal (Section 5.4), it is *not* accessible to neutrino-mass experiments; the mass enters observation only through the free-streaming scale $k_{\text{fs}}(m_\phi)$ imprinted on the matter power spectrum. The falsifiable channel is therefore the analytic $k_{\text{fs}} = 0.754\,h/\text{Mpc}$ (with a CLASS-measured half-mode at $k_{1/2} = 0.337\,h/\text{Mpc}$), probed by the Lyman-$\alpha$ forest and by DESI Y3 / Euclid clustering (2026–2028); see prediction 3.

2. $\sigma_8^{\text{eff}} = 0.747$ and $S_8 = 0.758$: the two-sector CLASS run gives a specific prediction for the amplitude of matter fluctuations. This S_8 sits at 0.04σ from KiDS (0.759 ± 0.024), *resolving* the lensing tension at the linear level. If future Euclid weak-lensing constraints confirm $S_8 < 0.74$ or $S_8 > 0.80$, the two-sector model is falsified.

3. $f\sigma_8$ **at** 0.93σ: the two-sector $f\sigma_8$ prediction is testable with DESI Y5 RSD measurements (2026–2027). A confirmed $f\sigma_8 > 0.47$ at $z = 0.15$ would tension the model at $> 1\sigma$.

4. **Background conserved**: $H_0 = 67.96$, $(w_0, w_a) = (-0.840, -0.670)$, and $\Delta\text{BIC}_{\text{CMB}} = -23.9$ remain as in Papers 1–5. Any experiment detecting $H_0 > 70$ or $w_0 > -0.7$ falsifies SSEE-V3.6 at the background level.

8.7 Status of the parameter count in the two-sector extension

The SSEE-V3.6 philosophy requires that no number be *fitted* to data. We distinguish here between *not fitted to data* (i.e., fixed prior to the comparison) and *derived from first principles*. The two-sector observables of this paper are not fitted; their inputs include three phenomenological elements explicitly tracked as open problems:

- $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\text{SSEE}} - 1)\,\Omega_{\text{m,dyn}}$: derived from the MIRA algebraic identity. *Not fitted.* (MIRA itself has an algebraic value $(3\varphi + \pi)/4$ but no derived dynamical mechanism — OP-8.)

- $m_\phi = \Sigma m_\nu^{\text{active}}\,(\text{SOLAR}^2 \cdot \text{KRYSTOS}) = 40.70\,\text{eV}$: a zero-fitting forward prediction. The multiplier $\text{SOLAR}^2 \cdot \text{KRYSTOS} = 594.28$ is a pure (φ, π) number and is the mass term of the written scalar Lagrangian (Section 8.5), so the mass is dimensionally consistent. *Not fitted.* What remains open is the UV origin of the multiplier (refined OP-9); $\Sigma m_\nu^{\text{active}}$ is Type P (OP-14).

- ξ: non-minimal coupling, currently treated as a free parameter (OP-11). Constrained below by the production mechanism.
- $\sigma_8^{\rm eff} = 0.747$ and $S_8 = 0.758$: CLASS outputs of the two-sector run (cold + ϕ-DM ncdm), with the suppression scale emerging from the Boltzmann hierarchy. *Not fitted* — S_8 lands at 0.04σ from KiDS as a forward prediction, not by tuning.

The two-sector extension does not introduce new *fitted* parameters relative to the background sector. The mass m_ϕ is no longer a loose ansatz: its coefficient is the mass term of a written scalar Lagrangian, which closes the dimensional-incompleteness of OP-9 and leaves only the UV origin of the multiplier $\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}$ open. The remaining phenomenological inputs are the non-minimal coupling ξ (OP-11) and the UV origin of the multiplier (refined OP-9). (The former MIRA matter mechanism, OP-8, is dissolved under the ω_m-direct reframe: there is no matter factor to derive.) Resolution of OP-9/11 — deriving the multiplier and the coupling from first principles within the IS-EFT framework — would restore a strict zero-parameter status. This is left for future work.

9 Conclusions

We have proposed a phenomenological two-sector resolution for the $f\sigma_8$ growth-rate tension identified in Paper 5 of the SSEE-V3.6 series. The key results are:

1. **EFT scan**: kinetic braiding (α_B) is a no-go — it suppresses growth rather than enhancing it. Run-rate modification ($\alpha_M = -0.08$) reduces the tension to 1.57σ but requires tuning to the observational bound.
2. **Two-sector model**: the MIRA algebraic identity $\mathcal{M}_{\rm SSEE} \approx 2$ implies a second dark matter sector, $\Omega_{\phi\text{-DM}} = \Omega_{\rm m,CMB} - \Omega_{\rm m,dyn} = 0.30889 - 0.160 = 0.14889$, with a free-streaming cutoff at $k_{\rm fs}$.
3. **Mass from a written Lagrangian**: $m_\phi = \Sigma m_\nu^{\rm active}\,(\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}) = 0.0685 \times 594.28 = 40.70$ eV, with the multiplier a pure (φ, π) number that is the mass term of the scalar Lagrangian $\mathcal{L}_\Phi = \frac{1}{2}(\partial\Phi)^2 - \frac{1}{2}m_\phi^2\Phi^2$ (Section 8.5). This is dimensionally consistent and closes the incompleteness of OP-9; what remains open is the UV origin of the multiplier (refined OP-9), with $\Sigma m_\nu^{\rm active}$ Type P (OP-14). Not fitted to growth data.
4. **Verification (class, zero fitting)**: the two-sector run gives $\sigma_8^{\rm eff} = 0.747$ and $S_8 = 0.758$ (0.04σ KiDS, *resolving* the lensing tension vs. ΛCDM $\sim 3\sigma$) and $f\sigma_8$ mean tension $\mathbf{0.93}\sigma$ (single-sector baseline: 0.70σ; the free-streaming that resolves S_8 raises this modestly, still $< 1\sigma$, close to ΛCDM 0.73σ). The CMB acoustic peaks ($\ell = 219/533/807$) and RMS(TT) $= 0.93\%$ vs. ΛCDM are conserved because $m_\phi \approx 37$ eV is non-relativistic well before recombination. All background tests — BAO ($\chi_r^2 = 0.01$), clusters ($\chi_r^2 = 0.122$) — conserved.

The framework SSEE-V3.6-V3.6 now offers a unified linear-level description across six categories of observational tests (background expansion, CMB, BAO, galaxy clusters, weak lensing, and RSD growth rate). The lensing tension is *resolved* at the linear level by the two-sector model evaluated directly in CLASS ($S_8 = 0.758$ at 0.04σ from KiDS), with no warm-dark-matter transfer function fitted to data: the suppression scale is a Boltzmann output. The mass coupling $m_\phi = \Sigma m_\nu^{\rm active}$ (SOLAR^2 · KRYSTOS) is now the mass term of a written scalar Lagrangian; the free-streaming scale $k_{\rm fs}$ still awaits verification against Lyman-α forest data. The remaining background tension ($H(z)$, $\chi^2_r = 1.861$) is structural and traceable to $\Omega_{\rm m,dyn} = 0.160$.

The **falsifiable predictions** of this paper are: (1) the ϕ-dark matter free-streaming scale $k_{\rm fs} = 0.754\,h/{\rm Mpc}$ (with a CLASS-measured half-mode at $k_{1/2} = 0.337\,h/{\rm Mpc}$), set by the mass $m_\phi = 40.70\,{\rm eV}$ and probed by the Lyman-α forest and by DESI Y3 / Euclid galaxy clustering (2026–2028) — the relevant channel because ϕ-DM is a gravitationally produced scalar with no Standard-Model portal (Section 5.4), and so leaves no neutrino-sector signature; and (2) $\sigma_8^{\rm eff} = 0.747$ ($S_8 = 0.758$), testable by Euclid weak-lensing (2026–2028).

Data Availability

All scripts, data files, and compiled PDFs are available at https://github.com/mikealmeida1721/SSEE. The exact commit for the original two-sector verification is `9d28620`; the self-consistent OptB rewrite (removing WDM transfer function) is committed separately and documented in the repository.

Author Contributions

Single-author work. M.E.A.V. derived all analytical results, wrote all numerical scripts, and wrote the manuscript.

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